

# **CHE simulator user manual**

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|------------------------------|---|
| 1 INTRODUCTION               | 4 |
| 2 User interface             | 5 |
| 2.1 Connection section       | 5 |
| 2.2 Current position section | 6 |
| 2.3 Heartbeat section        | 6 |
| 2.4 Status section           | 6 |
| 2.5 Job data section         | 7 |
| 2.6 Actions window           | 7 |

## 1 INTRODUCTION

This manual instructs how to operate with the simulator application. The simulator offers simple interface to TOS system including these features:

- connecting to TOS system as client or server
- sending pick/place messages to TOS system
- sending heartbeat messages to TOS messages
- receiving job messages from TOS system
- actions window to follow message flow

Before using the simulator it has to be installed by executing corresponding installer. It is a window executable (KcCheSimulatorInstaller.exe) and it creates startup icon to desktop.

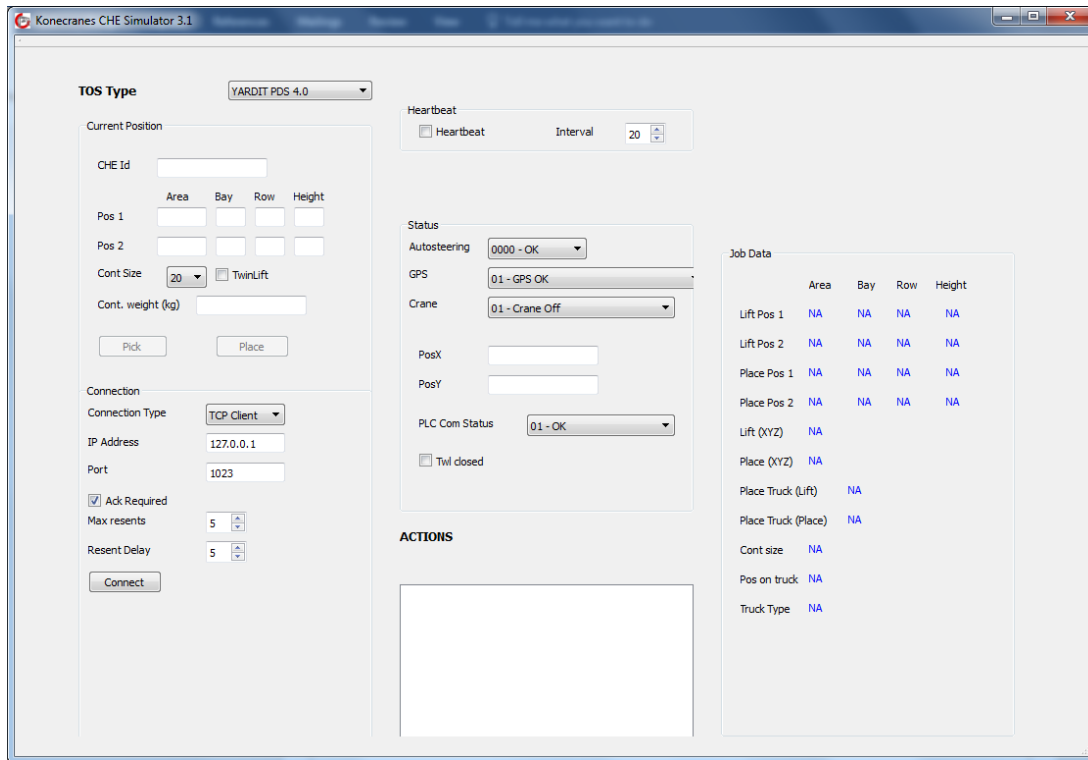
Icon that installer creates looks like this.



Start your application double clicking this icon.

## 2 User interface

User interface window looks like this



From **TOS Type** dropdown box you can select between three different versions

YARDIT PDS 4.0  
YARDIT PDS 3.3  
YARDIT PDS 3.0

Different sections of the user interface are handled in the next chapters.

### 2.1 Connection section

In **Connection Type** dropdown box you can select between TCP Client and TCP Server options. Other options are currently not working.

**IP Address** is the destination (TOS) computer IP address. If TCP Server is used, then address selection is not meaningful and is neglected because TCP server is listening at any port.

**Port** is the port where TOS system is listening (Client Mode) or where CHE simulator is listening (Server Mode).

If **Ack Required** check box is checked then job message is confirmed with acknowledge message. Also pick/place message expects acknowledgement.

**Max Resents** spin box used to set up how many times pick/place message is resent if not acknowledged. This setting is meaningful only if **Ack Required** check box is checked.

**Resent Delay** spin box is used to set up how many times pick/place message is resent if not acknowledged. This setting is meaningful only if **Ack Required** check box is checked.

**Connect/Listen** push button is used to connect to tcp server (client mode) or to start listening at predefined port (server mode).

## 2.2 Current position section

At **Cheld** text box the name of the simulated crane can be given.

At **Pos1** the position of the container 1 can be given.

At **Pos2** the position of container 2 (only twinlift) can be given.

Container size can be selected from dropdown list **Cont size**.

**Twinlift** can be activated by checking corresponding checkbox.

**Cont. Weight (kg)** text box can be used to feed in container weight in kilograms.

By pressing **Pick** or **Place** push buttons pick/place messages can be sent.

## 2.3 Heartbeat section

**Heartbeat** checkbox can be used to activate heartbeat message.

Heartbeat message will be sent at constant intervals defined at **Interval** selection box. Number defines the interval in seconds.

## 2.4 Status section

In this section the following data can be given to be sent in heartbeat message:

- Autosteering Status (OK/ERROR)
- GPS Status (selection between seven different statuses)
- Crane Status (selection between four different statuses)
- Crane X position in meters compared to base station
- Crane Y position in meters compared to base station.

- PLC communication status (OK/Failure)

## **2.5 Job data section**

In this section job data message information is displayed including

- Lift position 1 for container 1
- Lift position 2 for container 2 (only twinlift)
- Pick position 1 for container 1
- Pick position 2 for container 2 (only twinlift)
- Lift xyz position (if used)
- Pick xyz position (if used)
- Truck place for lift (if used)
- Truck place for pick (if used)
- Container size in feet
- Position on truck (if used)
- Truck Type (if used)

## **2.6 Actions window**

In actions window messages flowing from/to simulator will be displayed most recent message always at top.